ML PRACTICAL 4

Sales Data Clustering using K-Means

Aim

To group sales records into clusters using the K-Means clustering algorithm and visualize the resulting groups.

Theory

K-Means is an unsupervised machine learning algorithm that divides data into K groups (clusters) based on similarity. It assigns each data point to the nearest cluster centroid and repeatedly updates the centroids until the assignments become stable or the algorithm reaches its stopping condition.

Data Preparation

The notebook loads **sales_data_sample.csv** and selects four numerical features: **QUANTITYORDERED**, **PRICEEACH**, **SALES** and **MSRP**. StandardScaler is applied to scale the features so that variables with larger numerical ranges do not dominate the distance calculations.

Elbow Method

The Elbow Method is used to help choose a suitable number of clusters. The notebook fits K-Means models for K values from 1 to 8 and records their WCSS (Within-Cluster Sum of Squares). WCSS measures the sum of squared distances between data points and their assigned cluster centroids. The values are plotted against K; the bend or “elbow” can indicate a suitable cluster count.

K-Means Clustering

After examining the WCSS values, the notebook creates a final K-Means model with **3 clusters**. The model is fitted on the scaled features, and each sales record receives a cluster label. The number of records in each cluster is then displayed.

Visualization

A scatter plot is created using SALES on the X-axis and QUANTITYORDERED on the Y-axis. Different colours represent the cluster labels, helping visualize how the sales records are grouped.

Result

K-Means clustering is applied to the selected sales features. The notebook evaluates cluster counts using the Elbow Method, assigns records to 3 clusters and visualizes the groups.

Conclusion

This practical demonstrates how K-Means can identify groups in sales data without predefined labels. Feature scaling, the Elbow Method and cluster visualization support the clustering process and interpretation.

Theory is based on the uploaded ML_Prac_4 notebook. The exact elbow point and cluster counts should be taken from the executed notebook output.